

# Ahmed Ikram | AI Engineer

Dundee, Scotland | +44 7421 996048 | [ahmedikram30@gmail.com](mailto:ahmedikram30@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Online Portfolio](#)

Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee (predicted First, graduating June 2027). Built LLM systems with the evaluation harnesses that prove they work: RAGAS-scored retrieval, QLoRA and vLLM serving. Seeking a graduate AI engineering role from June 2027.

## EDUCATION

**University of Dundee - BSc (Hons) Computer Science (Data Science and AI)**

Sep 2023 - Present

- On track for a First-Class Honours degree. Expected Graduation: June 2027

**High School of Dundee**

Sep 2018 - June 2023

- Highers: A Mathematics, A Computing Science, A English, B Chemistry
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

## SKILLS

**Languages & Data:** Python · SQL · TypeScript · JavaScript · Bash · Rust / PyO3 · PostgreSQL · Redis · Kafka · Pydantic

**LLMs:** PyTorch · Hugging Face Transformers · QLoRA/LoRA/PEFT · Unsloth · TRL · Accelerate · BitsandBytes · vLLM · Jinja2, Prompt & Context Engineering

**Agents & RAG:** LangGraph · LangChain · Model Context Protocol (MCP) · AWS Bedrock · ChromaDB · Cross-Encoder Reranking · Multimodal (Vision-Language) Extraction · Guardrails

**Evaluation & Observability:** RAGAS · Execution-Based Scoring · LLM-as-Judge · LangSmith · Langfuse · Evidently AI

**MLOps & Serving:** MLflow · Weights & Biases · Apache Airflow · FastAPI · asyncio · GraphQL · Modal · AWS SageMaker

**Classic ML:** XGBoost · Isolation Forest · scikit-learn · Optuna · Tesseract · OpenCV

**Cloud:** AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager) · Google Cloud (GCS) · Terraform · Docker · GitHub Actions · Linux

## PROJECTS

### ATM Log Aggregation, Anomaly Detection & Diagnostics - RAG Assistant and ML Anomaly Detection (NCR Atleos)

LangGraph · MCP · ChromaDB · FastAPI · XGBoost · MLflow · Kafka · PostgreSQL · Redis · Docker · Terraform · AWS

- Hybrid RAG as a LangGraph agent over a 12-tool MCP server, semantic chunking & local `nomic-embed-text` embeddings, cross-encoder reranking, grounding-gated re-retrieval, then reflexion / self-consistency / citation checks fused with Platt-calibrated confidence (ECE gate 0.10).
- RAGAS evaluation across 3 retrieval strategies (50-query golden set): adaptive routing with re-retrieval nearly doubles context recall over deterministic selection (0.708 vs 0.350) at the best context precision (0.874); a CI gate blocks any merge regressing a RAGAS metric by >0.05.
- Three-layer architecture anomaly detection (XGBoost + Isolation Forest ensemble, Z-score & 7-heuristics engine) runs every 30s on a live Kafka Raft stream in a multi-host distributed architecture, reaching 0.70 F1 at 97.3% precision via a 200-point Youden sweep, versioned in MLflow with champion aliases and cross-checked against a live SageMaker endpoint.
- Terraform-provisioned AWS (ECS Fargate, RDS, S3, SageMaker, Secrets Manager) via OIDC-authenticated CI/CD, validated by 1,438 tests.

### StockLens FinTech Spending & Investment App - AI Agent, Multimodal Extraction, MLOps

LangGraph · MCP · PyTorch · Optuna · Rust / PyO3 · Airflow · MLflow · Evidently AI · Tesseract · GraphQL · Redis · Terraform

- 2-node LangGraph ReAct agent (StateGraph, not `create\_react\_agent` wrapper) exposes 16 tools to AWS Bedrock over a FastAPI-mounted Stateless MCP server (Streamable HTTP, OAuth 2.1 PKCE). Correctness judged in CI by an LLM-as-judge with LangSmith tracing.
- Confidence-gated receipt-extraction cascade (Tesseract OCR → Bedrock Vision-LLM → Text fallback) with rapidfuzz merchant matching, escalating only when confidence drops below thresholds.
- Global LSTM (entity embeddings, Optuna 50-trial HPO, focal loss) reaches 51.6% 5-day directional accuracy vs a 33% 3-class baseline (0.97 simulated Sharpe). 17 features computed in a Rust/PyO3 engine; retrained weekly via Airflow with Evidently drift monitoring (PSI/KS/JSD) and MLflow champion/challenger promotion at >2 pp.

- 2,413 tests with a 90% backend test coverage gate on Terraform-provisioned AWS (ECS Fargate, EFS champion serving); 9-parallel CI jobs and a 7-stage OIDC-authenticated CD.

### **SWE-Qwen Automated LLM Fine-Tuning, Evaluation & Serving**

*Python · PyTorch · QLoRA · Unsloth · Hugging Face · Modal · vLLM · Weights & Biases · Langfuse · Terraform · Google Cloud*

- Model-agnostic LLMOps platform architecture; schema-validated data preparation, QLoRA fine-tuning, execution-based evaluation and statistically-gated champion/challenger promotion, serving the winner as a config-driven OpenAI compatible API - versioned GCS/W&B artifacts shared across every stage.
- Six-stage data pipeline turned 20,477 raw SWE-bench issues into a leak-free 17,456-example corpus across 46 repos; evaluation materialises each repo in its official Docker image and runs its real pass/fail test suite for ground-truth, execution-based scoring.
- Scored 3 QLoRA Qwen3-14B variants (4-bit NF4, Unsloth + FlashAttention on Modal A100); champion reached 17.20% fail-to-pass - 7.0× base (95% CI 11.1-25.8%, McNemar  $p \approx 6e-05$ ) - and 90.10% unrelated pass-pass rate (+61.6 pt vs base); Promotion requires a 4-condition statistical gate so a challenger can never self-certify a win.
- Served on Modal scale-to-zero with per-request LoRA swapping via vLLM+AWQ; a 20-instance F2P smoke gate (SMOKE\_TOLERANCE=0.05) blocks regressing PRs. GCP via Terraform + Workload Identity Federation; 1,456 tests.

### **CERTIFICATES**

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|----------------------------------------------------------------------------|-----------------|
| <u>AWS Academy Graduate - Machine Learning Foundations</u>                 | <i>Feb 2025</i> |
| <u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u> | <i>Oct 2024</i> |
| <u>AWS Academy Graduate - Cloud Foundations</u>                            | <i>Mar 2024</i> |

### **EXPERIENCE**

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|-------------------------------------------------|-----------------------------|
| <b>Beaully Mini Market</b> , Dundee, Scotland   |                             |
| <i>Retail Assistant</i>                         | <i>Sep 2023 - Present</i>   |
| <b>Zulus Peri Peri Grill</b> , Dundee, Scotland |                             |
| <i>Front of House Worker</i>                    | <i>June 2021 - Sep 2023</i> |